



Clearance of hypermutator XDR *Pseudomonas* osteomyelitis and hardware infection with adjunctive bacteriophage therapy

Viseth Ngauy,¹ Scott Bevens,² Donna Kido,³ Rachel Ibrahimovic,³ Edwin Kamau,⁴ Angela Caldwell,⁴ Robert Pangilinan,⁵ Roy Levit,⁶ Francois Lebreton,⁷ Andrey A. Filippov,⁸ Ting L. Luo,⁷ Damon W. Ellison,⁹ Mikelson P. Nikolich,⁸ Patrick McGann⁷

AUTHOR AFFILIATIONS See affiliation list on p. 6.

ABSTRACT

Background Bacteriophages (phages), viruses ubiquitous in the environment infecting bacteria, are an increasingly attractive option for adjunct therapy with conventional antibiotics in complicated, refractory multidrug-resistant infections. Bacteria (“hypermutators”) with defects in methyl-directed mismatch repair (MMR) have been described in *Pseudomonas aeruginosa*, with the resultant defect in DNA repair responsible for rapid development of antimicrobial resistance.

Case Summary A 72-year-old male with mild chronic obstructive pulmonary disease, diagnosed with resectable, advanced right frontoethmoid adenoid cystic carcinoma, underwent endoscopic resection to remove the affected bone and tissue. The subsequent large craniofacial defect was reconstructed with non-vascularized calvarial bone grafts, rigidly fixated with titanium hardware and externally covered with a vascularized pericranial flap. He developed complete dehiscence of the skull base reconstruction, and wound cultures grew multiple susceptible *P. aeruginosa* isolates with distinct morphologies that became progressively drug resistant on treatment. Whole-genome sequencing of isolates showed the presence of a non-synonymous mutation in the *mutS* gene impacting the MMR pathway, consistent with a hypermutator. Phage therapy was pursued due to ongoing infection with retained hardware, lack of clinical response to antibiotic therapy alone, extreme antibiotic resistance, and possible surgical cure of his cancer. He received twice-daily intravenous phage therapy in combination with cefiderocol for 2 weeks with clinical and microbiological resolution of his infection.

Conclusion This case suggests that phage therapy may be useful in the treatment of deep-seated hypermutator *P. aeruginosa* infections with retained hardware.

KEYWORDS phage therapy, *Pseudomonas aeruginosa*, hypermutator

In the United States, phage therapy remains an investigational treatment available only through the Food and Drug Administration (FDA) under an Expanded Access (compassionate use) program. However, in 2022, the National Institute of Allergy and Infectious Diseases and the Antibacterial Resistance Leadership Group Phage Taskforce published a framework outlining potential uses for phage therapy in various infections and for treatment of multidrug-resistant (MDR) and/or biofilm-mediated infections (1).

Pseudomonas aeruginosa is one of the most challenging organisms to eradicate in chronic infections, primarily due to its intrinsic resistance to many antibiotics (2) and a tendency to colonize hospital sinks and drains (3). Furthermore, *P. aeruginosa* with hypermutator phenotypes is commonly seen in cystic fibrosis patients with chronic respiratory infections (4), with up to 30% of strains containing mutations (*mutS*, *mutL*,

Editor Carey-Ann D. Burnham, Pattern Bioscience, Austin, Texas, USA

Address correspondence to Viseth Ngauy, Viseth.ngauy.mil@health.mil.

A.A.F. and M.P.N. are party to a royalty-bearing material licensing agreement with Adaptive Phage Therapeutics, now merged with Biomx Inc.

See the funding table on p. 7.

Received 4 June 2025

Accepted 1 January 2026

Published 23 January 2026

[This article was published on 23 January 2026 with Fig. 1 in black and white. The figure was updated to color in the current version, posted on 4 February 2026.]

This is a work of the U.S. Government and is not subject to copyright protection in the United States. Foreign copyrights may apply.

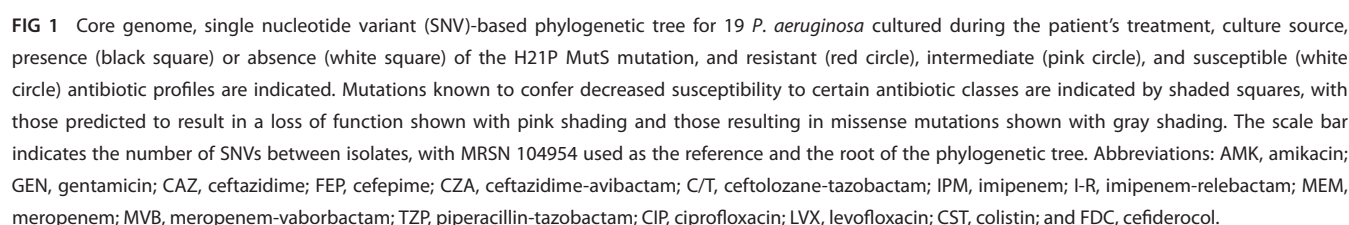
and *uvrD*) in DNA mismatch repair systems believed to potentiate a chronic, virulent milieu (5, 6).

CASE PRESENTATION

A 72-year-old male underwent resection of the frontal bone and sinuses, medial aspect of the orbit bone and contents with the adjacent nasoethmoidal sinus tract, and a large portion of the dura over the frontal lobe for advanced right frontoethmoid adenoid cystic carcinoma. The large composite defect was reconstructed with non-vascularized calvarial bone grafts, rigidly fixated with titanium hardware, and externally covered with a vascularized pericranial flap. His postoperative course was complicated by a right frontal lobe hemorrhage requiring prolonged intubation with eventual tracheostomy. Initial cultures grew three strains of pan-sensitive *P. aeruginosa* (Fig. 1, days 0 and 5) that subsequently became more resistant (Fig. 1, day 15). He was treated with broad-spectrum antibiotics for presumed ventilator-associated pneumonia and discharged to complete inpatient rehabilitation (Fig. 2A).

Three months after surgery, he developed complete dehiscence of the skull base reconstruction with external exposure of the bone and titanium implants. Cultures obtained from nasal discharge grew three distinct morphologies of *P. aeruginosa* (Fig. 1, day 105) susceptible to every antibiotic tested using the Vitek2 with Cards N808 and XN-32 (bioMérieux, France) with CLSI guidelines. MICs of cefiderocol (Shionogi Pharma, Japan) were determined in triplicate using broth microdilution per CLSI guidelines at concentrations ranging from ≤ 0.25 to 128 $\mu\text{g/L}$; all isolates had MICs $\leq 4 \mu\text{g/mL}$ (Fig. 1). He declined any surgical therapy and received oral ciprofloxacin for presumed osteomyelitis (~6 weeks) without improvement and ultimately underwent right orbital exenteration, debridement, and partial orbital hardware removal. The large cranial defect was repaired with a polyether ether ketone (PEEK) implant covered by a vascularized latissimus dorsi free flap. Histopathology of the resected right orbital apex showed margins free of malignancy. Cultures of intracranial tissue again grew *P. aeruginosa*, which was initially treated with ceftazidime/avibactam and tobramycin but switched to an extended course of meropenem based on antibiotic susceptibility testing (AST). He continued to show ongoing signs of infection with elevated inflammatory markers and leukocytosis after 4 weeks of therapy, prompting removal of the PEEK implant. Intraoperatively, an adjacent epidural abscess was discovered with two distinct morphologies of *P. aeruginosa* resistant to ciprofloxacin, ceftazidime, and cefepime isolated (Fig. 1, day 160). Superficial cultures of the exposed tissues grew multiple strains of *P. aeruginosa* with highly variable resistance profiles (Fig. 1, day 183–188). Multiple empiric regimens were started before transitioning to cefiderocol while awaiting susceptibility test results and drug availability. His clinical course and treatment regimen for the XDR *P. aeruginosa* are summarized in Fig. 2.

Notably, the patient used gentle nasal saline irrigations for mucosal dryness and to minimize discomfort from endoscopic procedures (last use before hospitalization on day 125). Culture of the content of the irrigation bottle grew *P. aeruginosa* with two distinct morphologies (Fig. 1, day N/A). All 19 strains (17 from the patient and 2 from the saline bottle) were sequenced as previously described (7) and showed that all belonged to the rare sequence type (ST) 782 and carried no acquired resistance genes. Notably, isolates differed by 0–243 single nucleotide variants (SNVs), with the first four isolates from days 0, 5, and 15 differing by just 0–3 SNVs (Fig. 1). The next isolate, cultured from nasal drainage on day 105, differed from the earlier isolates by 77–80 SNVs, an unexpectedly high number of SNVs for such a relatively short period. Further genomic analysis identified a non-synonymous mutation in the *mutS* gene resulting in a histidine to proline substitution at amino acid 21 (H21P) in the DNA repair protein *MutS*, a known method for acquiring a hypermutator phenotype (8). To further investigate the mutational resistome, 165 genes for which known chromosomal mutations have been linked to antibiotic resistance were investigated (9–14) using MRSN 104954 as reference. Twenty-eight distinct variants in 24/164 genes were identified in at least one patient



Phage selection and propagation

10.1128/asmcr.00108-25 **3**

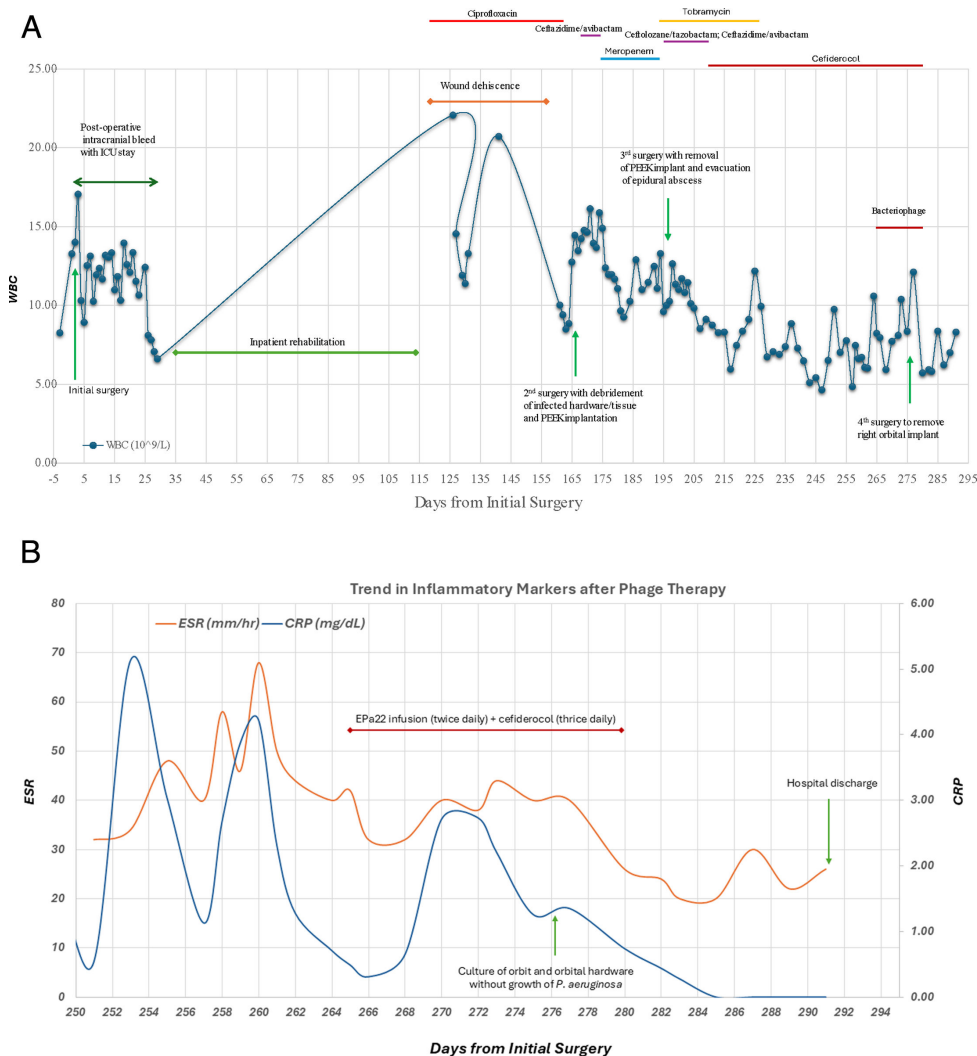


FIG 2 (A) Patient's antimicrobial treatment course against *P. aeruginosa* following initial surgery. Ongoing signs of infections with elevations in his white blood cell counts indicated incomplete or failed response to antibiotic therapy and surgery alone. (B) Response to phage-antibiotic combination therapy with normalization of erythrocyte sedimentation rate and C-reactive protein and microbiological clearance of *P. aeruginosa* on wound culture (day ~276).

DISCUSSION

Pseudomonas infections, particularly deep-seated infections such as osteomyelitis with implantable hardware, are difficult to treat due to the organism's propensity to produce biofilms and intrinsic resistance to many antibiotics (2). In this case, the pathogen was a hypermutator *P. aeruginosa*, likely acquired from a reusable, nasal saline spray bottle that was washed with tap water from his faucet. *P. aeruginosa* is a known pathogen found in water systems. Environmental surveillance of home carbonation systems, stainless steel and polyethylene terephthalate water bottles, and dishwasher seals has demonstrated significant bacterial load and colonization, along with other coliforms (20, 21).

Pirnay et al. reported eradication in 61.3% of infections in 100 European patients receiving personalized phage therapy when used in combination with antibiotics (22). Kohler et al. reported the successful treatment of a hypermutator MDR *P. aeruginosa* strain in a patient with Kartagener syndrome with repeated doses of aerosolized phage therapy in combination with intravenous antibiotic therapy for a chronic, life-threatening lung infection (23). We found only one other report of successful bacteriophage therapy

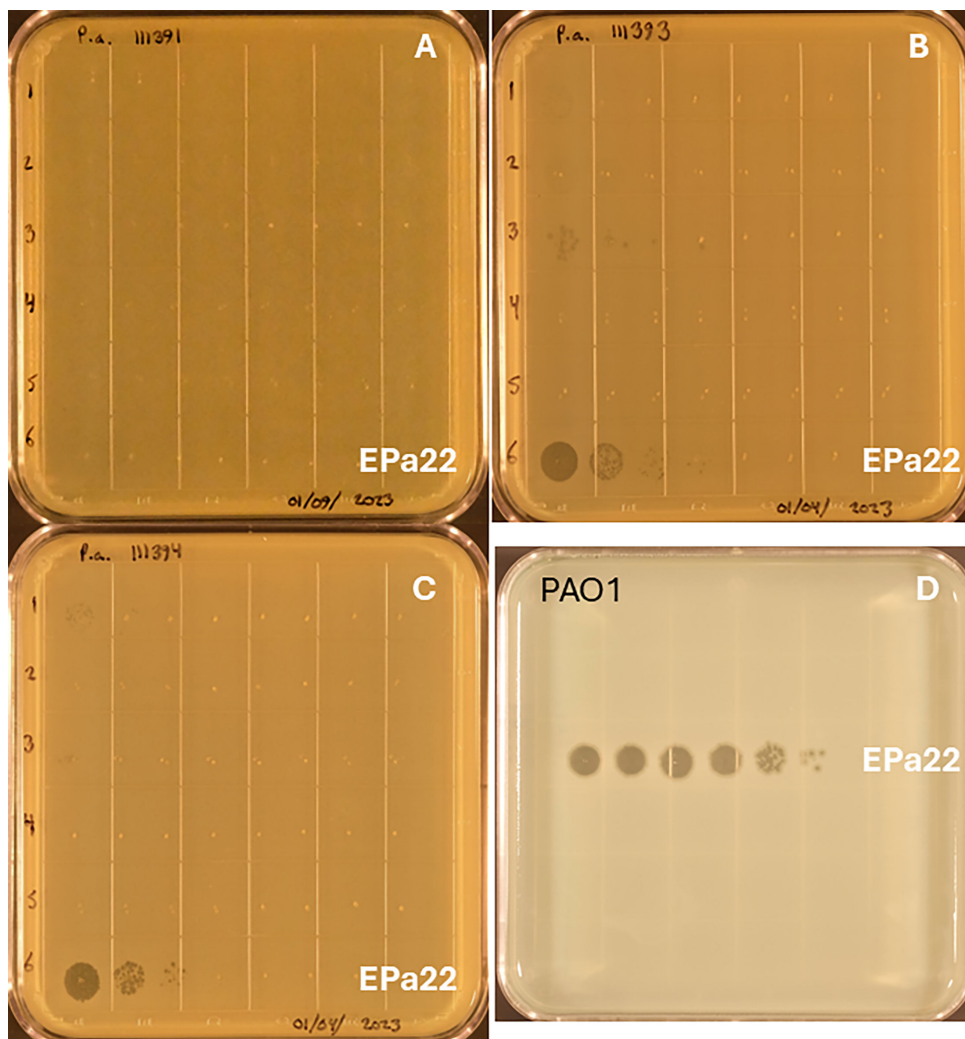


FIG 3 Susceptibility testing of several *P. aeruginosa* isolates from the patient to phage EPa22. Each spot consisted of 2 μ L of a 10-fold phage dilution applied to a semi-solid agar overlay infused with 0.1 mL of an overnight culture of the *P. aeruginosa* test strain, here MRSN 111391 (A), MRSN 111393 (B), and MRSN 111394 (C). As a positive control for lytic activity, an overnight culture of *P. aeruginosa* strain PAO1, a broadly permissive host with characterized phage susceptibility, was spotted with each tested phage (D). Tested as described previously (18), the patient's isolates exhibited varied susceptibility to 69 *P. aeruginosa* lytic phages from WRAIR's library of sequenced and phenotypically characterized therapeutic candidates (~150 unique phages). EPa22 was selected for intravenous therapy based on its robust lytic activity against all the patient isolates that were tested ($n = 9$) except MRSN 111391. Whole-genome sequence analysis of EPa22 (17) showed no bacterial gene content or evidence of a temperate lifestyle, indicating safety for therapeutic application. EPa22 underwent three rounds of single-plaque isolation and was propagated on strain PAO1, which is known to secrete exotoxin A (ETA) under iron-limiting growth conditions (19), but no ETA was detected by ELISA (Biomatix Corporation, Ontario, Canada). After propagation, lysates were purified by repeated centrifugation and filter-sterilization using sterile Millex GV 0.22- μ m syringe filters (SLGV033RS, EMD Millipore, Burlington, MA). Further purification was by cesium chloride fractionation and endotoxin removal using EndoTrap resin (Hyglos, Bernried am Starnberger See, Germany) followed by overnight dialysis against 1,000 $\times$ volumes of phosphate-buffered saline with 10 mM of magnesium sulfate (PBS-Mg). The target dose of EPa22 for intravenous administration was 1.0×10^9 PFU in 1.0 mL PBS-Mg; doses in this batch contain 141.6 EU, below the acceptable limit for human therapy of 5 EU per kilogram per hour (350 EU for a patient weighing 70 kg). Endotoxin content was tested using an Endosafe Nexgen-PTS spectrophotometer (Charles River Laboratories, Wilmington, MA). Sterility of single-dose vials and a bulk phage suspension (49 mL) of the purified EPa22 preparation was confirmed by testing per USP <71> at Focus Laboratories (<https://www.focus-lab.com>).

in combination with cefiderocol for treatment of cranial osteomyelitis caused by multiple, non-hypermutable, *P. aeruginosa* ST235 clones (24).

This is the first report of a successful treatment of a hypermutable strain of *P. aeruginosa* for osteomyelitis with retained hardware using intravenous bacteriophage therapy in combination with antibiotics in a non-cystic fibrosis (CF) patient. The logistics, administrative, and regulatory hurdles for using this phage were formidable but achievable (<2 months from FDA approval to administration). Our patient had several risk factors besides the extensively drug-resistant (XDR) *P. aeruginosa* infection that hampered his recovery; he was severely malnourished and debilitated from his malignancy, prolonged hospitalization, chronic infection, and poor appetite. He weighed just under 47 kg (down from 80 kg) at the time of his death. Additionally, this case highlights the risks associated with self-preparation of saline solutions and reuse of nasal bottles as adequate disinfection may not be achievable. Phage therapy may never garner the level of commercial interest or demand as antibiotics, but with limited and diminishing therapeutic options for MDR/XDR infections, it can be a useful adjunct. In this case, EPa22 appeared effective *in situ* despite a hypermutable phenotype producing multiple variants, suggesting the potential for use of phage in combination with antibiotics for retained hardware when removal carries significant morbidity.

Search strategy and selection criteria

We searched PubMed for relevant articles in English from database inception to 1 May 2025 for the following terms “hypermutable,” “hypermutable AND *Pseudomonas*,” “hypermutable AND bacteriophage,” “bacteriophage AND *Pseudomonas*,” “bacteriophage AND hardware,” and “bacteriophage AND osteomyelitis.” Each full article was manually reviewed for content and references pulled for additional information and used as references when applicable.

ACKNOWLEDGMENTS

We thank Dr. Robert Schooley and Ms. Kimberly McConnell for their expert advice; Dr. Olga Kirillina, Dr. Nino Mzhavia, Ms. Caitlin Urick, and the rest of WRAIR Wound Infections Department team for their work preparing and testing the EPa22 drug product; and the Tripler nursing, medical, pharmacy, and administrative staff for their assistance in this patient's care.

WRAIR phage susceptibility screening and lab manufacturing, purification, and testing of the EPa22 therapeutic preparation were supported by Congressionally Directed Medical Research Program grant PR182667.

V.N., P.M.G., M.P.N., and S.B. wrote the main manuscript text. P.M.G., F.L., and T.L.L. performed the WGS, data analysis, and figure preparation. V.N., R.P., S.B., R.I., R.L., and D.K. provided care and treatment for the patient and assisted with coordination and delivery of phage therapy. E.K. and A.C. provided microbiological support, testing, and analysis of the patient's clinical isolates. M.P.N., D.W.E., and A.A.F. selected, tested, and manufactured phage for use. All authors reviewed the manuscript.

The views expressed in this publication are those of the authors and do not necessarily reflect the official policy or position of the Defense Health Agency, Department of Defense, nor the U.S. Government.

AUTHOR AFFILIATIONS

¹Infectious Disease Service, Department of Medicine, Tripler Army Medical Center, Honolulu, Hawaii, USA

²Head and Neck Oncology, Otolaryngology Service, Tripler Army Medical Center, Honolulu, Hawaii, USA

³Department of Pharmacy, Tripler Army Medical Center, Honolulu, Hawaii, USA

⁴Department of Pathology and Area Laboratory Services, Tripler Army Medical Center, Honolulu, Hawaii, USA

⁵Department of Medicine, Tripler Army Medical Center, Honolulu, Hawaii, USA

⁶Internal Medicine Residency Program, Department of Medicine, Tripler Army Medical Center, Honolulu, Hawaii, USA

⁷Multidrug-Resistant Organism Repository and Surveillance Network (MRSN), Diagnostics and Countermeasures Branch, Center for Infectious Diseases Research (CIDR), Walter Reed Army Institute of Research, Silver Spring, Maryland, USA

⁸Wound Infections Department, Bacterial Diseases Branch, CIDR, Walter Reed Army Institute of Research, Silver Spring, Maryland, USA

⁹Bacterial Diseases Branch, CIDR, Walter Reed Army Institute of Research, Silver Spring, Maryland, USA

AUTHOR ORCIDS

Viseth Ngauy  <http://orcid.org/0009-0001-8239-3543>

Edwin Kamau  <https://orcid.org/0000-0002-5761-7883>

Francois Lebreton  <https://orcid.org/0000-0002-7157-5026>

Andrey A. Filippov  <https://orcid.org/0000-0003-2256-861X>

Mikeljon P. Nikolich  <https://orcid.org/0000-0001-5770-8980>

Patrick McGann  <https://orcid.org/0000-0003-1548-9438>

FUNDING

Funder	Grant(s)	Author(s)
Congressionally Directed Medical Research Programs	PR182667	Mikeljon P. Nikolich Andrey A. Filippov

DATA AVAILABILITY

The genomes of all strains sequenced in this project have been deposited at NCBI under Bioproject accession no. [PRJNA1354456](https://www.ncbi.nlm.nih.gov/bioproject/1354456).

ETHICS APPROVAL

Approval for use of the experimental phage therapy was obtained from the US Food and Drug Administration as a Single Patient Investigational New Drug (SPIND #29299) request. The Madigan Army Medical Center Institutional Review Board (IRB) review for the treatment protocol could not be obtained prior to product administration; however, concurrence of two independent physicians was obtained for the experimental therapy. The phage product was given under emergency use with written consent from the patient. All side effects were reported to the IRB within 5 business days of product administration in accordance with 21 CFR 56.104(c).

REFERENCES

- Suh GA, Lodise TP, Tamma PD, Knisely JM, Alexander J, Aslam S, Barton KD, Bizzell E, Totten KMC, Campbell JL, Chan BK, Cunningham SA, Goodman KE, et al. 2022. Considerations for the use of phage therapy in clinical practice. *Antimicrob Agents Chemother* 66:e0207121. <https://doi.org/10.1128/AAC.02071-21>
- Poole K. 2011. *Pseudomonas aeruginosa*: resistance to the max. *Front Microbiol* 2:65. <https://doi.org/10.3389/fmicb.2011.00065>
- Kizny Gordon AE, Mathers AJ, Cheong EYL, Gottlieb T, Kotay S, Walker AS, Peto TEA, Crook DW, Stoesser N. 2017. The hospital water environment as a reservoir for carbapenem-resistant organisms causing hospital-acquired infections—a systematic review of the literature. *Clin Infect Dis* 64:1435–1444. <https://doi.org/10.1093/cid/cix132>
- Hall KM, Pursell ZF, Morici LA. 2022. The role of the *Pseudomonas aeruginosa* hypermutator phenotype on the shift from acute to chronic virulence during respiratory infection. *Front Cell Infect Microbiol* 12:943346. <https://doi.org/10.3389/fcimb.2022.943346>
- Oliver A, Baquero F, Blázquez J. 2002. The mismatch repair system (mutS, mutL and uvrD genes) in *Pseudomonas aeruginosa*: molecular characterization of naturally occurring mutants. *Mol Microbiol* 43:1641–1650. <https://doi.org/10.1046/j.1365-2958.2002.02855.x>
- Oliver A, Cantón R, Campo P, Baquero F, Blázquez J. 2000. High frequency of hypermutable *Pseudomonas aeruginosa* in cystic fibrosis lung infection. *Science* 288:1251–1254. <https://doi.org/10.1126/science.288.5469.1251>
- Bartlett KV, Luo TL, Ong AC, Maybank RA, Stribling W, Thompson B, Powell A, Kwak YI, Bennett JW, Lebreton F, Mc Gann PT. 2024. Tn4661-mediated transfer of bla CTX-M-15 from *Klebsiella michiganensis* to an outbreak clone of *Pseudomonas aeruginosa*. *Microb Genom* 10. <https://doi.org/10.1099/mgen.0.001303>
- Dulanto Chiang A, Patil PP, Beka L, Youn JH, Launay A, Bonomo RA, Khil PP, Dekker JP. 2022. Hypermutator strains of *Pseudomonas aeruginosa* reveal novel pathways of resistance to combinations of cephalosporin antibiotics and beta-lactamase inhibitors. *PLoS Biol* 20:e3001878. <https://doi.org/10.1371/journal.pbio.3001878>
- Cabot G, López-Causapé C, Ocampo-Sosa AA, Sommer LM, Domínguez MÁ, Zamorano L, Juan C, Tubau F, Rodríguez C, Moyà B, Peña C,

- Martínez-Martínez L, Plesiat P, Oliver A. 2016. Deciphering the resistome of the widespread *Pseudomonas aeruginosa* sequence type 175 international high-risk clone through whole-genome sequencing. *Antimicrob Agents Chemother* 60:7415–7423. <https://doi.org/10.1128/AAC.01720-16>
10. Domitrovic TN, Hujer AM, Perez F, Marshall SH, Hujer KM, Woc-Colburn LE, Parta M, Bonomo RA. 2016. Multidrug resistant *Pseudomonas aeruginosa* causing prosthetic valve endocarditis: a genetic-based chronicle of evolving antibiotic resistance. *Open Forum Infect Dis* 3:ofw188. <https://doi.org/10.1093/ofid/ofw188>
 11. López-Causapé C, Cabot G, Del Barrio-Tofiño E, Oliver A. 2018. The versatile mutational resistome of *Pseudomonas aeruginosa*. *Front Microbiol* 9:685. <https://doi.org/10.3389/fmicb.2018.00685>
 12. López-Causapé C, Sommer LM, Cabot G, Rubio R, Ocampo-Sosa AA, Johansen HK, Figuerola J, Cantón R, Kidd TJ, Molin S, Oliver A. 2017. Evolution of the *Pseudomonas aeruginosa* mutational resistome in an international cystic fibrosis clone. *Sci Rep* 7:5555. <https://doi.org/10.1038/s41598-017-05621-5>
 13. Sherrard LJ, Tai AS, Wee BA, Ramsay KA, Kidd TJ, Ben Zakour NL, Whitley DM, Beatson SA, Bell SC. 2017. Within-host whole genome analysis of an antibiotic resistant *Pseudomonas aeruginosa* strain sub-type in cystic fibrosis. *PLoS One* 12:e0172179. <https://doi.org/10.1371/journal.pone.0172179>
 14. Sobel ML, Neshat S, Poole K. 2005. Mutations in PA2491 (mexS) promote MexT-dependent mexEF-oprN expression and multidrug resistance in a clinical strain of *Pseudomonas aeruginosa*. *J Bacteriol* 187:1246–1253. <https://doi.org/10.1128/JB.187.4.1246-1253.2005>
 15. Aslam S, Roach D, Nikolich MP, Biswas B, Schooley RT, Lilly-Bishop KA, Rice GK, Cer RZ, Hamilton T, Henry M, et al. 2024. *Pseudomonas aeruginosa* ventricular assist device infections: findings from ineffective phage therapies in five cases. *Antimicrob Agents Chemother* 68:e0172823. <https://doi.org/10.1128/aac.01728-23>
 16. Engeman E, Freyberger HR, Corey BW, Ward AM, He Y, Nikolich MP, Filippov AA, Tyner SD, Jacobs AC. 2021. Synergistic killing and re-sensitization of *Pseudomonas aeruginosa* to antibiotics by phage-antibiotic combination treatment. *Pharmaceuticals (Basel)* 14:184. <https://doi.org/10.3390/ph14030184>
 17. Farlow J, Freyberger HR, He Y, Ward AM, Rutvisuttinunt W, Li T, Campbell R, Jacobs AC, Nikolich MP, Filippov AA. 2020. Complete genome sequences of 10 phages lytic against multidrug-resistant *Pseudomonas aeruginosa*. *Microbiol Resour Announc* 9:e00503-20. <https://doi.org/10.1128/MRA.00503-20>
 18. Sergueev KV, Filippov AA, Farlow J, Su W, Kvachadze L, Balarjishvili N, Kutateladze M, Nikolich MP. 2019. Correlation of host range expansion of therapeutic bacteriophage Sb-1 with allele state at a hypervariable repeat locus. *Appl Environ Microbiol* 85:e01209-19. <https://doi.org/10.1128/AEM.01209-19>
 19. Ochsner UA, Johnson Z, Lamont IL, Cunliffe HE, Vasil ML. 1996. Exotoxin A production in *Pseudomonas aeruginosa* requires the iron-regulated pvdS gene encoding an alternative sigma factor. *Mol Microbiol* 21:1019–1028. <https://doi.org/10.1046/j.1365-2958.1996.481425.x>
 20. Hariharan AV, Sankar MM. 2024. Daily use water bottles as a hub for microbial population: a comparative study of PET vs. stainless steel water bottles and outcome of washing strategy intervention. *J Pharm Bioallied Sci* 16:S1242–S1245. https://doi.org/10.4103/jpbs.jpbs_559_23
 21. Zupančič J, Turk M, Črnigoj M, Ambrožič Avguštin J, Gunde-Cimerman N. 2019. The dishwasher rubber seal acts as a reservoir of bacteria in the home environment. *BMC Microbiol* 19:300. <https://doi.org/10.1186/s12866-019-1674-5>
 22. Pirnay J-P, Djebara S, Steurs G, Griselain J, Cochez C, De Soir S, Glonti T, Spiessens A, Vanden Berghe E, Green S, et al. 2024. Personalized bacteriophage therapy outcomes for 100 consecutive cases: a multicentre, multinational, retrospective observational study. *Nat Microbiol* 9:1434–1453. <https://doi.org/10.1038/s41564-024-01705-x>
 23. Köhler T, Luscher A, Falconnet L, Resch G, McBride R, Mai Q-A, Simonin JL, Chanson M, Maco B, Galiotto R, Riat A, Civic N, Docquier M, McCallin S, Chan B, van Delden C. 2023. Personalized aerosolised bacteriophage treatment of a chronic lung infection due to multidrug-resistant *Pseudomonas aeruginosa*. *Nat Commun* 14:3629. <https://doi.org/10.1038/s41467-023-39370-z>
 24. Simner PJ, Cherian J, Suh GA, Bergman Y, Beisken S, Fackler J, Lee M, Hopkins RJ, Tamma PD. 2022. Combination of phage therapy and cefiderocol to successfully treat *Pseudomonas aeruginosa* cranial osteomyelitis. *JAC Antimicrob Resist* 4:dlac046. <https://doi.org/10.1093/acamr/dlac046>